CHAPTER 13

MOVEMENTS OF OCEAN WATER

CHAPTER ESSENCE AT A GLANCE:
Master ocean waves and orbital water kinematics, gravitational dynamics of spring and neap tides, tidal bores, planetary driving forces of ocean currents, global subtropical gyres, and thermohaline circulation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Wave Mechanics and Orbital Dynamics:** Ocean surface waves are oscillations generated by wind friction; water particles execute circular orbital trajectories in place while the wave waveform and energy propagate horizontally.

• **Tidal Gravitational Pull and Alignment:** Tides are periodic rises and falls in ocean sea level caused by gravitational attraction of the Moon and Sun; Spring tides occur during syzygy (full moon and new moon, highest tidal range), while Neap tides occur during quadrature (quarter moon, lowest tidal range).

• **Primary and Secondary Drivers of Ocean Currents:** Primary drivers include solar insolation heating, wind shear friction, gravity, and Coriolis deflection; secondary drivers include density and salinity gradients and coastal boundary topography.

• **Global Gyres and Marine Thermohaline Currents:** Currents circulate in vast closed gyres (clockwise in Northern Hemisphere, counter-clockwise in Southern Hemisphere); warm currents (Gulf Stream, Kuroshio) elevate coastal warmth, while cold currents (California, Benguela) desiccate adjacent coastlines.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$c = \sqrt{\frac{g\lambda}{2\pi}} \rightarrow \text{Deep Water Ocean Wave Celerity and Dispersion}$$

Deep-Water Surface Gravity Wave Propagation Velocity

$$F_{\text{tide}} \propto \frac{GM_{\text{body}}R_{\text{earth}}}{D^3} \rightarrow \text{Moon Tidal Pull is } \approx 2.17 \times \text{ Sun}$$

Inverse Cube Differential Gravitational Tidal Forcing

$$\text{Tidal Range Duality} \rightarrow \text{Spring Tide } (M+S) \longleftrightarrow \text{Neap Tide } (M-S)$$

Syzygy Alignment and Quadrature Minimum Tidal Range

$$\theta_{\text{net}} = 90^\circ \text{ (Right in North, Left in South)} \rightarrow \text{Ekman Mass Transport Deflection}$$

Net Ekman Ocean Water Mass Transport Heading

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The Moon exerts more than double (2.17 times) the tide-generating force of the Sun on Earth's oceans because tidal force varies inversely with the cube of distance (D3), overcoming the Sun's immense mass.

• **Alert:** Spring tides have nothing to do with the spring season; they occur twice every month on new moon and full moon days when the Sun, Moon, and Earth align in a straight line (syzygy).

• **Alert:** Where warm and cold ocean currents converge (such as the Gulf Stream and cold Labrador Current off Newfoundland), dense navigation-disrupting fog is produced, but the area becomes a world-class fishing bank.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Spring Tides and Neap Tides based on the relative astronomical positions of the Earth, Moon, and Sun. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Spring Tides:** Occur when the Sun, Moon, and Earth are in a straight line (syzygy) during new moon and full moon; gravitational forces reinforce each other, producing the highest high tides and lowest low tides. **[1 Mark]**

• **Neap Tides:** Occur when the Sun and Moon are at right angles to each other (quadrature) with respect to Earth during quarter moon; gravitational forces counteract each other, resulting in the lowest tidal range. **[1 Mark]**

2. What is a Tidal Bore? Give one notable example of a river where a tidal bore occurs. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A steep, wall-like surge of water formed when an incoming high tide enters a funnel-shaped, shallow river estuary against the natural downstream river current. **[1 Mark]**

• **Example:** The Hooghly River in West Bengal (India) or the Qiantang River in China. **[1 Mark]**

3. Explain how wind generates ocean waves and define Wavelength, Wave Height, and Wave Period. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Wave Generation:** Wind blowing across the sea exerts friction on surface water, transferring energy and creating rhythmic orbital undulating disturbances. **[1 Mark]**

• **Wavelength and Wave Height:** Wavelength is the horizontal distance between two successive wave crests or troughs; Wave height is the vertical distance between the crest and trough. **[1 Mark]**

• **Wave Period:** The time interval required for two consecutive wave crests to pass a fixed stationary observation point. **[1 Mark]**

4. Differentiate between Warm Ocean Currents and Cold Ocean Currents with two examples of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Warm Ocean Currents:** Currents flowing from warm equatorial regions towards colder polar latitudes, raising the temperature of adjacent coastal areas (e.g. Gulf Stream in Atlantic, Kuroshio Current in Pacific). **[1.5 Marks]**

• **Cold Ocean Currents:** Currents flowing from polar and high latitudes towards equatorial warm regions, lowering coastal temperatures and creating coastal fog (e.g. Labrador Current in Atlantic, California Current in Pacific). **[1.5 Marks]**

5. Explain the major primary and secondary forces that generate and influence the circulation of ocean currents. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Heating by Solar Energy:** Unequal solar heating causes thermal expansion of water at the equator; equatorial water is about 8 cm higher than polar water, creating a slight gradient slope. **[1 Mark]**

• **Planetary Wind Friction:** Prevailing planetary trade winds and westerlies drag surface ocean water in their general wind direction. **[1 Mark]**

• **Coriolis Force:** Deflects ocean currents to their right in the Northern Hemisphere (forming clockwise gyres) and to their left in the Southern Hemisphere (forming counter-clockwise gyres). **[1 Mark]**

• **Salinity and Density Differences:** Denser, highly saline water sinks and flows as deep bottom currents, driving global thermohaline conveyor circulation. **[1 Mark]**

6. Discuss the major ecological and economic impacts of ocean currents on coastal climate, navigation, and commercial fisheries. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Coastal Climate Modification:** Warm currents like the North Atlantic Drift keep ports of north-western Europe ice-free even in deep winter; cold currents induce extreme aridity in adjacent west-coast deserts (e.g. Atacama, Namib). **[1.5 Marks]**

• **World Class Fisheries:** Zones where warm and cold currents mix produce ideal temperature and upwelling of rich mineral nutrients, supporting massive plankton growth and world-renowned fisheries (e.g. Grand Banks off Newfoundland). **[1.5 Marks]**

• **Maritime Navigation:** Ships travel faster when sailing with the current, saving fuel and transit time; however, convergence zones create dangerous dense fog and drifting icebergs. **[1 Mark]**